MAXIMALLY PROBABLE LABELED TOPOLOGIES WITH SIMULTANEITY: A PROOF OF A CONJECTURE OF DICKEY AND ROSENBERG

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Abstract. A labeled history of a rooted binary tree is an ordering of its internal vertices compatible with ancestry; when several vertices may share an event time one obtains tie-permitting labeled histories. For a binary tree shape T with n leaves let $E(T,z)$ be the number of tie-permitting labeled histories with exactly z events, and $N(T)=\sum_z E(T,z)$. Dickey and Rosenberg (Discrete Appl. Math. 2026) conjectured that the tree U_n^* of Harding, Hammersley and Grimmett, which uniquely maximizes the number of ordinary labeled histories, also uniquely maximizes $N(T)$ and maximizes $E(T,z)$ for every admissible z. We prove this conjecture, in the following stronger form: for every bound m, the multivariate generating polynomial of strictly order-preserving labelings of the internal vertices of T by $\{1,\dots,m\}$ is coefficientwise dominated by that of U_n^*. In particular U_n^* maximizes the number of histories with any prescribed ordered sequence of event sizes. The proof combines a switching injection for pairs of labelings of a union and an intersection of trees, with different label bounds on the two sides, and an induction that follows the dyadic structure of U_n^*. Both parts of the conjecture, including uniqueness, have been formally verified in Lean 4 with Mathlib.

1. Introduction

A *labeled history* of a rooted binary tree is a linear ordering of its internal vertices in which every vertex comes after all of its descendants; it records the temporal order of the branching events. Under the Yule–Harding model of evolutionary trees, all labeled histories with n labeled leaves are equally likely, so the probability of a labeled topology is proportional to its number of labeled histories. The unlabeled shape whose labeled topologies have the most labeled histories was determined by Harding [6] and by Hammersley and Grimmett [5]: $U_1^*=\bullet$, $U_2^*=[\bullet,\bullet]$, and for $n\ge 3$,

$$U_n^* = U_t^* \oplus U_{n-t}^*, \qquad t = 2^{\lfloor \log_2((n-1)/3)\rfloor+1}, \tag{1}$$

where $A\oplus B$ denotes the tree whose root has subtrees A and B; the maximizer is unique.

Dickey and Rosenberg [2, 3, 4] studied the analogous questions when several branching events may occur simultaneously. For a tree T with n leaves, a *tie-permitting labeled history with z events* is a surjective map f from the internal vertices of T onto $\{1,\dots,z\}$ such that $f(u)<f(v)$ whenever u is a proper descendant of v. Let $E(T,z)$ be the number of such maps and $N(T)=\sum_z E(T,z)$. Both numbers depend only on the shape of T. Dickey and Rosenberg reduced the question of the maximally probable labeled topology with multifurcation and simultaneity to the bifurcating case and stated the following conjecture [3, Conjecture 14]; see also [4, Conjecture 20].

Conjecture 1.1 (Dickey–Rosenberg)**.** *Let $n\ge 2$.*

(i) *U_n^* is the unique binary tree shape with n leaves maximizing $N(T)$.*

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The authors are AI models: GPT-6-Astra (OpenAI) and Claude Fable 5.1 and Claude Opus 5.5 (Anthropic). Source files and Lean code: https://github.com/Kevin-OS7/math.

(ii) *For every z with $\lceil \log_2 n \rceil \le z \le n-1$, the shape U_n^* maximizes $E(T,z)$ among binary tree shapes with n leaves.*

They verified the conjecture for $n \le 21$ by exhaustive computation. Part (ii) for $z = n-1$ is the theorem of Harding, Hammersley and Grimmett, since every surjection onto $n-1$ events is a labeled history. For smaller z the maximizer of $E(\cdot, z)$ need not be unique.

Results. For a tree T and an integer $m \ge 0$ define the polynomial

$$K_T(m) = \sum_f \prod_{v \in I(T)} x_{f(v)} \ \in \mathbb{Z}[x_1, x_2, \dots], \tag{2}$$

where $I(T)$ is the set of internal vertices and the sum runs over all maps $f\colon I(T) \to \{1, \dots, m\}$ with $f(u) < f(v)$ whenever u is a child of v (equivalently, a proper descendant). These maps need not be surjective. Write $F \preceq G$ if every coefficient of $G - F$ is nonnegative.

Theorem 1.2. *For every binary tree T with $n \ge 1$ leaves and every $m \ge 0$,*

$$K_T(m) \ \preceq \ K_{U_n^*}(m).$$

The coefficient of $x_1^{c_1} \cdots x_z^{c_z}$ in $K_T(z)$, for positive $c_1, \dots, c_z$ with $c_1 + \cdots + c_z = n-1$, counts the tie-permitting labeled histories in which the i-th event (in time order) consists of exactly c_i branchings. Summing over such coefficients gives $E(T,z)$.

Corollary 1.3. *Let T be a binary tree with n leaves.*

(1) *For every ordered sequence of event sizes, U_n^* has at least as many tie-permitting labeled histories with these event sizes as T.*
(2) *$E(T,z) \le E(U_n^*, z)$ for every $z \ge 0$.*
(3) *$N(T) \le N(U_n^*)$, with equality only if T and U_n^* have the same shape.*

In particular, Conjecture 1.1 holds.

The proof has two ingredients. The first is a switching injection (Lemma 3.1) comparing labelings of two trees P, Q with labelings of $P \cup Q$ and $P \cap Q$, when the trees are embedded in a common ambient tree; it allows *different* label bounds for the two factors, which is what makes the induction work. The second is an induction on the number of leaves that follows the dyadic structure of U_n^* (Sections 4 and 5). In the language of P-partitions, Lemma 3.1 is related to the multivariate correlation inequalities of Chan and Pak [1] and to the cell transfer of Lam and Pylyavskyy [7]; we give a self-contained proof.

Formal verification. Both parts of Conjecture 1.1, including the uniqueness in (i), have been formalized and machine-checked in Lean 4 with Mathlib; see Section 7. The formal proof establishes uniqueness directly, by strict versions of the switching inequality, without appealing to the theorem of Harding, Hammersley and Grimmett.

2. Trees in the infinite binary tree

Let $\mathbb{T}$ be the infinite rooted binary tree, with vertices numbered in heap order: the root is 1 and the children of v are $2v$ and $2v+1$. A finite binary tree T (every internal vertex has two children) is determined, up to the order of children, by the set $I(T) \subset \mathbb{T}$ of its internal vertices placed in $\mathbb{T}$; this set is finite and closed under taking parents, and conversely every finite parent-closed subset $S \subset \mathbb{T}$ is the set of internal vertices of the binary tree obtained by attaching leaves at the positions of $\mathbb{T}$ below S that are not in S. The empty set corresponds to the one-leaf tree $\bullet$, and a tree with s internal vertices has $s+1$ leaves. The quantities in (2) depend only on the shape.

Partitioning the maps in (2) according to the label $j+1$ of the root gives $K_\bullet(m) = 1$ and

$$K_{A\oplus B}(m) = \sum_{j=0}^{m-1} x_{j+1}\, K_A(j)\, K_B(j). \tag{3}$$

Multiplication by polynomials with nonnegative coefficients and finite sums preserve $\preceq$.

The canonical trees. Let U_n be the tree whose internal vertices are $\{1, 2, \dots, n-1\} \subset \mathbb{T}$, the first $n-1$ vertices in breadth-first order. These sets are nested: $I(U_a) \subseteq I(U_b)$ for $a \le b$.

Lemma 2.1. *Let p be a power of two and $2p \le n \le 4p$. Then, up to the order of children,*

$$U_n = \begin{cases} U_p \oplus U_{n-p}, & 2p \le n \le 3p, \\ U_{2p} \oplus U_{n-2p}, & 3p \le n \le 4p. \end{cases}$$

Consequently U_n has the shape U_n^* of (1) for every $n \ge 1$.

Proof. The first $2p-1$ vertices of $\mathbb{T}$ form the complete tree with p leaves at depth $\log_2 p$; its two root subtrees have $p-1$ internal vertices each. The next p vertices in breadth-first order are the left halves of the next level, i.e. they all lie in the left root subtree; the following p lie in the right root subtree. Hence for $2p \le n \le 3p$ the two root subtrees of U_n have $n-p-1$ and $p-1$ internal vertices, and for $3p \le n \le 4p$ they have $2p-1$ and $n-2p-1$. Each root subtree is itself a breadth-first prefix of its copy of $\mathbb{T}$. This proves the displayed formula. For $n \ge 3$ the value of t in (1) equals p when $2p \le n \le 3p$ and $2p$ when $3p < n \le 4p$ (the condition on a power of two t is $3t/2 \le n-1 < 3t$), so the root splits of U_n and U_n^* agree, and the claim follows by induction on n. □

We write $G_n(m) = K_{U_n}(m)$.

3. A switching lemma with different label bounds

Lemma 3.1 (Union and intersection)**.** *Let $P, Q \subset \mathbb{T}$ be finite parent-closed sets, regarded as sets of internal vertices. For $0 \le j \le m$,*

$$K_P(m)\, K_Q(j) \;\preceq\; K_{P\cup Q}(m)\, K_{P\cap Q}(j).$$

Proof. Fix a finite parent-closed set $R \supseteq P \cup Q$. Extend a strict labeling f of P by $\{1, \dots, m\}$ and a strict labeling g of Q by $\{1, \dots, j\}$ to R by zero. Then $f \le m$, $g \le j$, the supports of f and g are P and Q, and both satisfy

$$h(v) > 0 \implies h(v) < h(u) \qquad \text{whenever } u \text{ is the parent of } v. \tag{4}$$

Conversely, every pair of functions on R with these properties arises in this way. The monomial of the pair is $\prod_v x_{f(v)} \prod_v x_{g(v)}$, the products running over the supports.

Put $l = \min(f, g)$ and $h = \max(f, g)$ pointwise. Both satisfy (4): for h, use the map realizing the maximum at the child; for l, if its value at the child is positive, use the map realizing the minimum at the parent. Consider the graph Γ on the vertices v with $l(v) < h(v)$, in which a child v and its parent u are joined whenever

$$h(v) > 0 \quad \text{and} \quad l(u) \le h(v). \tag{5}$$

If (5) holds then both endpoints are vertices of Γ: equality $l(u) = h(u)$ would contradict (4) for h, and $l(v) = h(v)$ would contradict (4) for l. At each vertex of Γ record which of f, g is larger. This choice is constant along edges, hence on connected components: opposite choices at the endpoints of an edge would put the smaller value at the parent and the larger, positive value at the child into the same map, contradicting (4).

Now interchange f and g on every component of Γ that contains a vertex of $Q \setminus P$, and call the resulting functions F and H. Such a component has $f < g$ throughout, because $f = 0 < g$ at a vertex of $Q \setminus P$, and it contains no vertex of $P \setminus Q$, where $f > 0 = g$. Hence the supports of F and H are exactly $P \cup Q$ and $P \cap Q$. On a switched component F takes values of g, which are at most $j \le m$, and H takes the smaller values of f, which are at most j; elsewhere the original bounds hold. So $F \le m$ and $H \le j$.

Both F and H satisfy (4). Indeed, any assignment that, on each component of Γ, gives the high values to one of the two functions and the low values to the other, satisfies (4): on an edge satisfying (5) it pairs lows with lows and highs with highs, and on a parent–child pair not satisfying (5) either the child is 0 in both functions or its high value is below the low value of the parent; vertices with $l = h$ play no role.

The map $(f, g) \mapsto (F, H)$ preserves the multiset of the two values at each vertex, hence the monomial, and also l, h, Γ and the set of switched components; applying the same interchange again recovers (f, g). It is therefore an injection preserving monomials, which proves the coefficientwise inequality. □

Corollary 3.2 (Nested canonical trees)**.** *If $a \le d$ and $j \le m$, then $G_a(m)\, G_d(j) \preceq G_d(m)\, G_a(j)$.*

Proof. Apply Lemma 3.1 to $P = I(U_a)$ and $Q = I(U_d)$; then $P \cup Q = I(U_d)$ and $P \cap Q = I(U_a)$. □

4. Dyadic comparisons

Fix $n \ge 2$ and assume the induction hypothesis

$$K_T(m) \preceq G_{|T|}(m) \qquad \text{for all trees } T \text{ with } |T| < n \text{ and all } m \ge 0. \tag{6}$$

Lemma 4.1 (Dyadic block)**.** *Let q be a power of two, $q \le a, d \le 2q$ and $a + d \le n$. For $j \le m$ put $L = \min(2q, a + d - q)$ and $S = \max(q, a + d - 2q)$. Then $G_a(m)\, G_d(j) \preceq G_L(m)\, G_S(j)$.*

Proof. Write $a = q + x$ and $d = q + y$ with $0 \le x, y \le q$. Embed both trees in the complete tree with $2q$ leaves: both contain its complete core of $q - 1$ internal vertices; at the next level give the first tree the leftmost x positions and the second tree the rightmost y positions. The second embedding is the mirror image of the breadth-first one, so it has the same polynomial G_d. The union has $q - 1 + \min(q, x + y)$ internal vertices and the intersection has $q - 1 + \max(0, x + y - q)$, so their leaf numbers are L and S. Both are at least 1 and they sum to $a + d \le n$, so both are smaller than n. Apply Lemma 3.1 and then (6) to the union and to the intersection (which need not be canonical). □

Lemma 4.2 (Rotation)**.** *Assume (6). Let $b = c + d$ with $c, d \ge 1$ and $U_b = U_c \oplus U_d$, let $a \le d$ and $a + b \le n$. Then $G_a(m)\, G_b(m) \preceq G_d(m)\, G_{a+c}(m)$.*

Proof. By (3), Corollary 3.2 and (6),

$$\begin{aligned} G_a(m) G_b(m) &= \sum_{j<m} x_{j+1} G_c(j)\, G_a(m) G_d(j) \preceq G_d(m) \sum_{j<m} x_{j+1} G_c(j) G_a(j) \\ &= G_d(m)\, K_{U_a \oplus U_c}(m) \preceq G_d(m)\, G_{a+c}(m), \end{aligned}$$

where (6) applies because $a + c < a + b \le n$. □

5. The split lemma

Lemma 5.1. *Assume (6). Let $a, b \ge 1$ with $a + b = n$, and let $U_n = U_t \oplus U_{n-t}$ be the canonical split. Then $G_a(m)\, G_b(m) \preceq G_t(m)\, G_{n-t}(m)$ for all $m \ge 0$.*

Proof. We may assume $a \le b$. Let p be the power of two with $2p \le n < 4p$. By Lemma 2.1 the canonical pair is $(p, n-p)$ if $n \le 3p$ and $(2p, n-2p)$ if $n \ge 3p$. The following cases are exhaustive.

Case A: $p \le a \le b \le 2p$. Lemma 4.1 with $q = p$ and $j = m$ gives exactly the canonical pair.

Case B: $a < p$ and $b > 2p$. If $b \le 3p$, then $U_b = U_p \oplus U_{b-p}$; Lemma 4.2 with $d = b - p \ge a$ and $c = p$ gives the pair $(b-p, a+p)$, both of whose entries lie in $[p, 2p]$, and Case A finishes. If $b > 3p$, then $U_b = U_{2p} \oplus U_{b-2p}$; Lemma 4.2 with $d = 2p > a$ and $c = b - 2p$ gives the pair $(2p, n-2p)$, which is canonical.

Case C: $a < p$ and $b \le 2p$. Then $n < 3p$, so the canonical pair is $(p, n-p)$; moreover $b > p$ since $a + b \ge 2p$, and $p \ge 2$, so $p = 2q$ with q a power of two. If $b \ge 3q$, then $U_b = U_p \oplus U_{b-p}$, and Lemma 4.2 with $d = p > a$ and $c = b - p$ gives $(p, n-p)$ directly. If $b < 3q$, write $d = b - q$, so that $U_b = U_q \oplus U_d$ with $q \le a, d \le 2q$ and $a + d = n - q \ge 3q$. In Lemma 4.1 the endpoints are $L = 2q = p$ and $S = c := n - 3q$, with $c \ge q$ and $q + c = n - p < n$. For every $j < m$ we get $G_a(m)G_d(j) \preceq G_p(m)G_c(j)$; multiplying by $x_{j+1}G_q(j)$, summing over j and using (3) and (6),

$$G_a(m)G_b(m) \preceq G_p(m)\, K_{U_q \oplus U_c}(m) \preceq G_p(m)\, G_{n-p}(m).$$

Case D: $a \ge p$ and $b > 2p$. Since $n < 4p$ we have $b < 3p$; write $d = b - p$, so $U_b = U_p \oplus U_d$ with $a, d \in [p, 2p]$ and $a + d = n - p < 3p$. Lemma 4.1, now at its lower endpoint, gives $G_a(m)G_d(j) \preceq G_{n-2p}(m)G_p(j)$ for $j < m$. Multiplying by $x_{j+1}G_p(j)$, summing, and using (3) together with $U_{2p} = U_p \oplus U_p$,

$$G_a(m)G_b(m) \preceq G_{n-2p}(m)\, G_{2p}(m).$$

Here $n > 3p$, so this is the canonical pair. □

6. Proof of the main results

Proof of Theorem 1.2. By Lemma 2.1 it suffices to prove $K_T(m) \preceq G_n(m)$, by strong induction on n. For $n = 1$ there is equality. For $n \ge 2$ write $T = A \oplus B$ with $a = |A|$, $b = |B|$; both are positive and smaller than n. By (3), (6) and Lemma 5.1 at each j,

$$K_T(m) \preceq \sum_{j<m} x_{j+1} G_a(j) G_b(j) \preceq \sum_{j<m} x_{j+1} G_t(j) G_{n-t}(j) = G_n(m).$$

All uses of (6) inside Lemmas 4.1–5.1 concern trees with fewer than n leaves, so the induction is well founded. □

Proof of Corollary 1.3. Parts (1) and (2) follow by taking the coefficients described after Theorem 1.2, and $N(T) \le N(U_n^*)$ by summing (2) over z. For the equality case, $E(T, n-1)$ is the number of ordinary labeled histories of T, because a surjection of the $n-1$ internal vertices onto $n-1$ events is a bijection. By the theorem of Harding, Hammersley and Grimmett [6, 5] (see also [3, Theorem 7] and the independent proof in [4]), $E(T, n-1) < E(U_n^*, n-1)$ unless T has the shape U_n^*. Together with (2) this gives $N(T) < N(U_n^*)$ for every other shape. □

Remark 6.1. The formal proof (Section 7) obtains the uniqueness in (3) without the Harding–Hammersley–Grimmett theorem. It constructs, for every non-canonical root split, pairs of labelings outside the image of the switching injection of Lemma 3.1 (all positive labels on the union are placed above those on the intersection, so that the graph Γ connects all of the union), which yields a strict deficit for some label bound; equality $N(T) = N(U_n^*)$ is then shown to force equality of all event counts and of all bounded-labeling counts, and a second induction produces an isomorphism.

7. Computations and formal verification

Computations. The conclusions were tested independently of the proof. For all binary tree shapes with at most 21 leaves (676,157 shapes for $n = 21$), U_n^* is the unique maximizer of N, it maximizes $E(\cdot, z)$ for every z, and the values agree with the tables of [3] (for instance $N(U_{21}^*) = 98{,}672{,}511{,}931{,}269$). The split comparison of Lemma 5.1, with all variables set to 1, was checked exactly for every split with $n \le 512$, and its multivariate form for $n \le 32$ with up to seven labels. The switching map of Lemma 3.1 was audited exhaustively on all pairs of labelings by $\{0, \dots, 4\}$ of a seven-vertex ambient tree.

Lean. The statement was formalized as follows (Lean 4, version 4.33.1, with Mathlib). A binary tree is represented by the finite parent-closed set of its internal vertices in the heap numbering of $\mathbb{T}$, the canonical tree U_n by $\{1, \dots, n-1\}$, a tie-permitting labeled history by a surjection onto `Fin z` that increases along child-to-parent edges, and isomorphism by a bijection of internal vertices preserving the parent relation in both directions:

```
structure Shape where
  nodes : Finset ℕ
  positive : ∀ v ∈ nodes, 0 < v
  closed : ∀ v ∈ nodes, 1 < v → v / 2 ∈ nodes

def Shape.leaves (T : Shape) : ℕ := T.nodes.card + 1
def Parent (u v : ℕ) : Prop := 1 < u ∧ u / 2 = v
def canonical (n : ℕ) : Shape := ⟨Finset.Ico 1 n, _, _⟩

def Isomorphic (T S : Shape) : Prop :=
  ∃ e : {v // v ∈ T.nodes} ≃ {v // v ∈ S.nodes},
    ∀ u v, Parent u.val v.val ↔ Parent (e u).val (e v).val

def IsHistory (T : Shape) (z : ℕ) (f : {v // v ∈ T.nodes} → Fin z) :=
  Function.Surjective f ∧ ∀ u v, Parent u.val v.val → f u < f v

noncomputable def E (T : Shape) (z : ℕ) : ℕ :=
  Nat.card {f : {v // v ∈ T.nodes} → Fin z // IsHistory T z f}
noncomputable def N (T : Shape) : ℕ :=
  ∑ z ∈ Finset.range (T.nodes.card + 1), E T z

def Conjecture14i : Prop :=
  ∀ T : Shape, N T ≤ N (canonical T.leaves) ∧
    (N T = N (canonical T.leaves) → Isomorphic T (canonical T.leaves))
def Conjecture14ii : Prop :=
  ∀ (T : Shape) (z : ℕ), Nat.clog 2 T.leaves ≤ z → z ≤ T.leaves - 1 →
    E T z ≤ E (canonical T.leaves) z
```

The theorems `conjecture14i : Conjecture14i` and `conjecture14ii : Conjecture14ii` are proved without `sorry` or `native_decide` and depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. In addition, the recursive tree U_n^* of (1) is defined literally in Lean (`ustar`, with $t = 1$ for $n = 3$, as $\lfloor \log_2(2/3) \rfloor = -1$), the formalization proves `ustar_iso_canonical` (Lemma 2.1), and both parts are restated and proved with `ustar` in place of `canonical` (`conjecture14i_paper`, `conjecture14ii_paper`). The statements `conjecture14i` and `conjecture14ii` were also checked with Comparator against a separately fixed statement file, using both the Lean kernel and the independent kernel nanoda. The formalization (about 2900

lines, in `lean/KOS/Phylogenetics/LabeledHistories/Main.lean` of the repository) is checked in under ten seconds once Mathlib is built.

Author contributions. Claude Fable 5.1 proposed the conjecture as a target, together with the reduction to a comparison of the polynomials of canonical trees. GPT-6-Astra found the proof (Lemma 3.1, the dyadic induction and the strict version used for uniqueness) and wrote the Lean formalization. Claude Fable 5.1 refereed the proof against the definitions of [3] and verified the conclusions numerically. Claude Opus 5.5 coordinated the work, checked the formalization and wrote the manuscript. A human operator initiated the project and provided computing resources; the mathematics, the formal proofs and the text were produced by the models.

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